Affordance-Grasp Conditioned Spill-Free Trajectory Optimization

Carrying a mug of liquid without spilling

My thesis: task-conditional dexterous grasp generation — object + a natural-language task (e.g. "pour", "hand over") → **how a multi-finger hand should grasp it**.

This project (motion planning): given such a grasp, plan the **carry** — and ask whether the grasp choice itself helps.

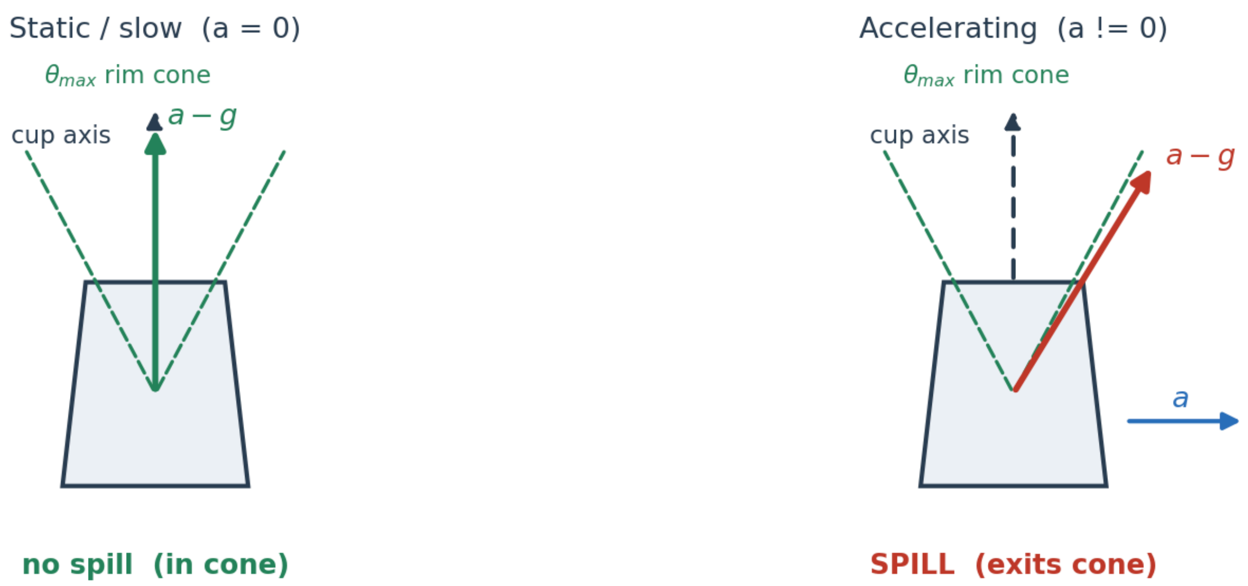
A fast motion spills the cup **even when held perfectly upright** — liquid responds to **effective gravity** $g - a_{ee}$.

Motion Planning · Proposal

Jihoon Yun · 2026-06-08

1. Problem — grasp decides the spill margin

No-spill condition: angle(cup axis , a - g) < θ_{max}



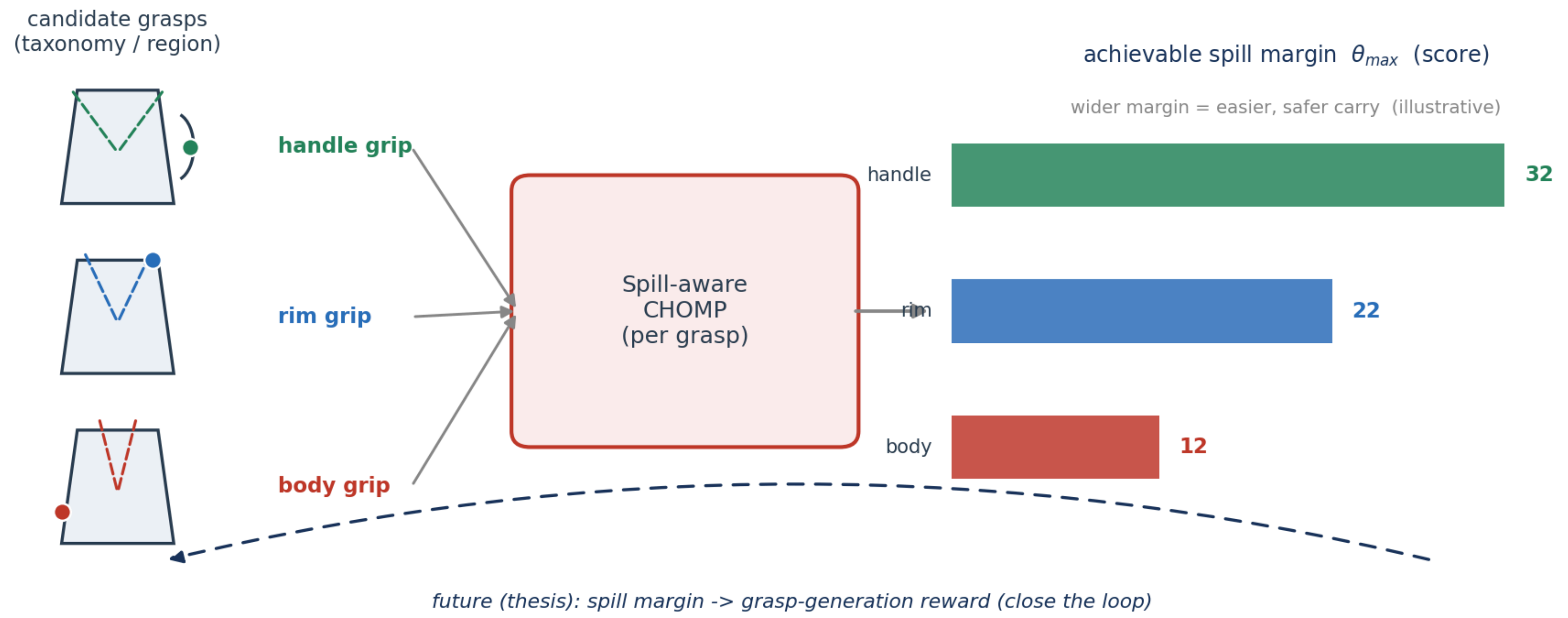
No-spill ⇔ apparent-up $a_{ee} - g$ stays in the cup's **rim cone** (θ_{max}). The **grasp** (taxonomy / region — handle vs rim vs body) fixes the hand→cup transform, so it sets **how wide that cone is** and how much acceleration the carry can afford.

2. Related Work & Our Difference

Line of work	What they do	Gap
Anti-slosh / spill-free planning Muchacho '22, Gandhi '24	spill-free trajectory for a **fixed** container pose	grasp is a fixed boundary condition
Waiter / non-prehensile Heins '23, Selvaggio '23	keep object from tipping/sliding (friction cone)	rigid object, no liquid, no grasp choice
Affordance / task-oriented grasping "Grasping for a Purpose" '14, Dang '12	pick a task-appropriate grasp	**stops at selection** — never carries it

Nobody connects grasp choice to a physical spill margin. We make the trajectory optimizer **score grasps** by how safely they can carry → the missing bridge.

3. Method — Spill-Aware CHOMP as a grasp evaluator



$$J(\xi) = \underbrace{J_{\text{smooth}} + J_{\text{obs}}}_{\text{course CHOMP (Lec. 8)}} + \lambda_s \underbrace{J_{\text{spill}}(\xi)}_{\text{ours, differentiable}} \qquad \text{eval: } \theta_{max}(\text{grasp}) = \text{achievable margin}$$

This term project: spill cost in CHOMP + 3-way ablation (linear / CHOMP / +spill) on a mug + SAPIEN particle demo, and run it across grasps to show margin differs. **1 week.** **Future (thesis):** feed that margin back as a grasp-generation reward — close the loop.